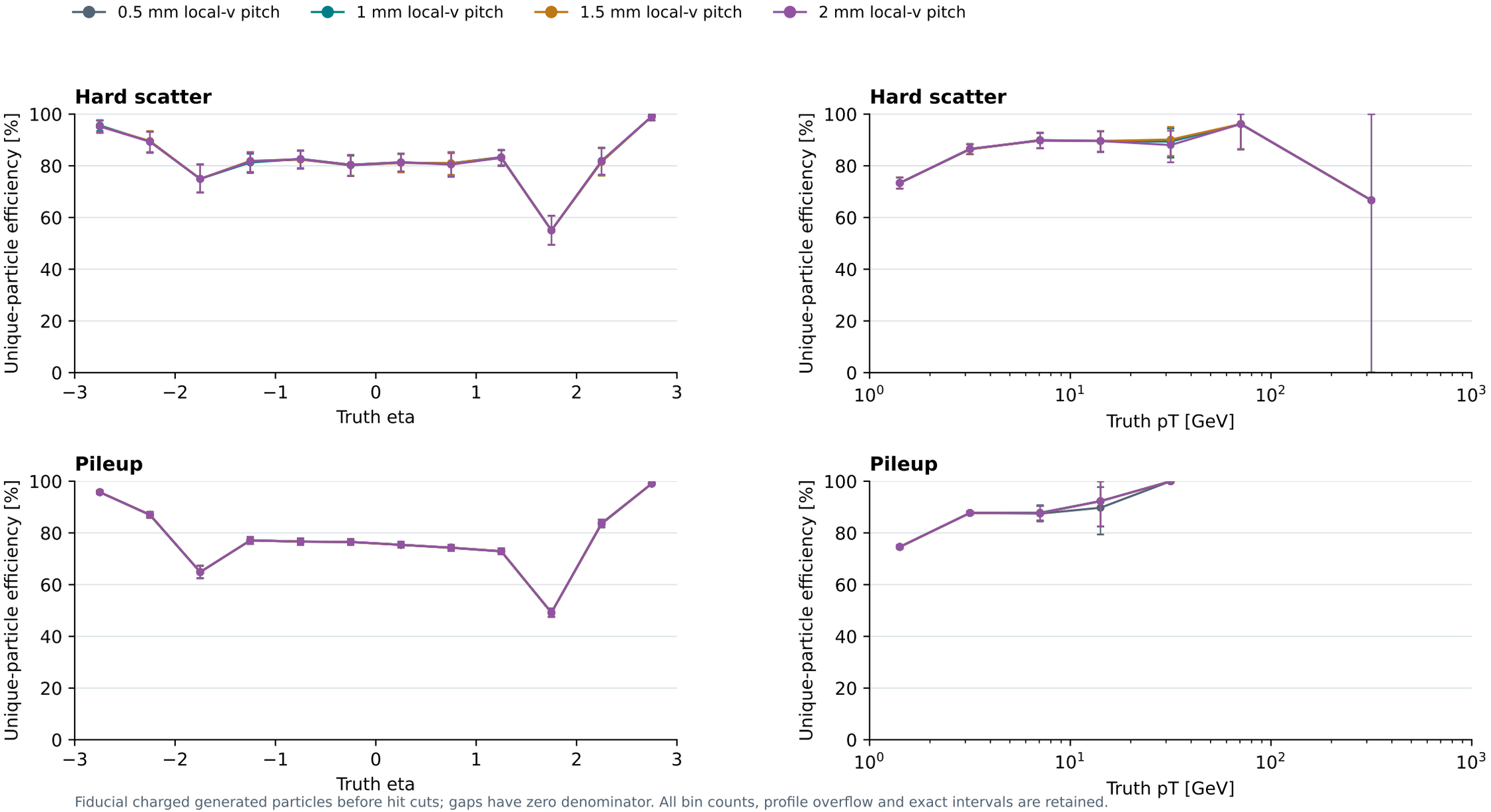
# Efficiency versus truth kinematics

Fatras qqbar-only ttbar + 200 pileup · 100 paired events/length · frozen selection · GBTS z0 [-1000, 1000] mm




Fiducial charged generated particles before hit cuts; gaps have zero denominator. All bin counts, profile overflow and exact intervals are retained.

Pointwise 95% paired event-block bootstrap. Constant intervals can be degenerate; they do not establish equivalence.

Scaled local-v Gaussian response, fixed material. Boundary retries may change non-target noise and seeds; see pairing audit.

PROTOTYPE · balanced tracker · constant 3 T · measured-pixel GBTS + full-sensor CKF + frozen ambiguity resolution

GBTS graph edge limit 2,000,000 reached - 0.5 mm: 99/100 events; 1 mm: 99/100 events; 1.5 mm: 99/100 events; 2 mm: 99/100 events. Resource-limited chain.